TEST-3

(a) Simple Random Walk

A drunk person is standing at a lamp post labelled 0 on an infinite street with streetlights labelled by all integers $\mathbb{Z}$

p with right forward step

$q = 1-p$ with left backward step

$\{X_n, n \geq 0\}$

State space $S = \{0, \pm 1, \pm 2, \dots\}$

Index set $T = \{0, 1, 2, 3, \dots\}$

$P(X_n = 1) = P(\text{Right}) = p$

$P(X_n = -1) = P(\text{Left}) = 1 - p = q$

Transition Probabilities:

$$P(X_{n+1} = j \mid X_n = i) = P(X_{n+1} = j - i) = \begin{cases} p, & j = i+1 \\ q, & j = i-1 \\ 0, & \text{otherwise} \end{cases}$$

Birth - Death Process

Consider a small bank customers:

$\lambda_i = q_{i,i+1} \equiv$ Arrival rate

$\mu_i = q_{i,i-1} \equiv$ leave rate

$T_i^A \equiv$ Time until an arrival occurs

$T_i^L \equiv$ Time until a leave occurs

$\{X(t), t \geq 0\}$

State space $S = \{0, 1, 2, 3, \dots\}$

Transition Rates:

$\to i+1 : q_{i,i+1} = \lambda$

$\to i-1 : q_{i,i-1} = \mu$

$i \sim \text{Exp}(\lambda + \mu)$

$t \in [0, \infty)$

(b) 2-state DTMC:

→ $X_n = 0$: it is raining on day n

→ $X_n = 1$: It is not raining on day n

$$X_n = \begin{cases} 0 \\ 1 \end{cases}$$

State space: $\{0,1\}$

Index set $T = \{0,1,2,\dots\}$

Transition probabilities:

$P_{ij} = P(X_{n+1} = j \mid X_n = i)$. $i,j \in \{0,1\}$

$P_{00} = P(X_{n+1} = 0 \mid X_n = 0)$

$P_{01} = P(X_{n+1} = 1 \mid X_n = 0)$

$P_{10} = P(X_{n+1} = 0 \mid X_n = 0)$

$$P_{ij} = P = \begin{bmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{bmatrix}$$

2-state CTMC

$X(t) = 0$: raining at time t

$X(t) = 1$: not raining at time t

$t \in [0,\infty)$, $S = \{0,1\}$

- fi = on exponential holding in each state.

Transition Rates:

$q_{01} = \lambda \quad q_{10} = \mu$

$V_i = \sum_{j \neq i} q_{ij}$

$V_0 = q_{01} + q_{0-1} = q_{01} = \lambda$

$V_1 = q_{12} + q_{10} = q_{10} = \mu$

$P_{ij} = \frac{q_{ij}}{V_i}, j \neq i$: $\quad P_{01} = \frac{q_{01}}{V_0} = \frac{\lambda}{\lambda} = 1$

$P_{10} = \frac{q_{10}}{V_1} = \frac{\mu}{\mu} = 1$

Transition probabilities to

$P_{ij}(t) = P(X(t) = j \mid X(0) = i)$

(c) Embedded discrete time Markov chain of a CTMC:

Consider a birth & death process $\{X(t), t \geq 0\}$.

State space $S = \{0,1,2\}$

→ $i+1$ at birth rate $\lambda_i \simeq q_{i,i+1}$

→ $i-1$ at death rate $\mu_i \simeq q_{i,i-1}$

$V_i = q_{i,i+1} + q_{i,i-1} = \lambda_i + \mu_i$

$P_{ij} = \frac{q_{ij}}{V_i}, \; j = i-1, i+1$

Embedded DTMC:

$P_{i,i+1} = \frac{q_{i,i+1}}{V_i} = \frac{\lambda_i}{\lambda_i + \mu_i}$

$P_{i,i-1} = \frac{q_{i,i-1}}{V_i} = \frac{\mu_i}{\lambda_i + \mu_i}$

$P_{ij} = 0$, if $|i-j| > 1$

Transition Probabilities:

$P_{ij}(t) = P(X(t) = j \mid X(0) = i)$

$= P(X(t) = j \mid X(0) = i)$

Question 2

Let $X(t)$ be the number of infected people at time t.

$\{X(t), t \geq 0\}$

a) State space $S = \{0, 1, 2, 3, \ldots\}$

Index set $t \in [0, \infty) \to$ Continuous-time process.

b) Transition Probabilities for the next Jump

- $i+1$: either due to infection or migration
- $i-1$: due to recovery

$q_{i,i+1} = \beta i + \sigma$

Infection: βi

Migration: σ

$q_{i,i-1} = \alpha i$

Recovery: αi

Total rate of leaving state i: $v_i = q_{i,i+1} + q_{i,i-1} = \beta i + \sigma + \alpha i = i(\beta+\alpha)+\sigma$

$$P_{i,i+1} = \frac{\beta i + \sigma}{i(\beta+\alpha)+\sigma}$$

$$P_{i,i-1} = \frac{\alpha i}{i(\beta+\alpha)+\sigma}$$

c) Transition Probabilities for Infinitesimal Time $h \to 0$

$$P(X(t+h) = j \mid X(t) = i) = \begin{cases} q_{ij} \cdot h + o(h), & \text{if } j \neq i \\ 1 - v_i \cdot h + o(h), & \text{if } j = i \end{cases}$$

$$P(X(t+h) = i+1 \mid X(t) = i) = (\beta i + \sigma)h + o(h)$$

$$P(X(t+h) = i-1 \mid X(t) = i) = (\alpha i)h + o(h)$$

$$P(X(t+h) = i \mid X(t) = i) = 1 - (\beta i + \sigma + \alpha i)h + o(h)$$

$$P_{ij}(h) = \begin{cases} (\beta i + \sigma)h + o(h) \\ (\alpha i)h + o(h) \\ 1 - (\beta i + \sigma + \alpha i)h + o(h) \\ o(h); \ |j-i| > 1 \end{cases}$$

d) $X(0) = 5$, $\beta = 0.02$, $\alpha = 0.03$, $\sigma = 0.01$

$$P_{56} \cdot P_{65} \cdot P_{54} = \frac{5(\beta)+\sigma}{5(\beta)+\sigma+5(\alpha)} \cdot \frac{6(\alpha)}{6(\beta)+\sigma+6(\alpha)} \cdot \frac{5(\alpha)}{5(\beta)+\sigma+5(\alpha)}$$

$$= \frac{5(0.02)+0.01}{5(0.02)+0.01+5(0.03)} \cdot \frac{6(0.03)}{6(0.02)+0.01+6(0.03)} \cdot \frac{5(0.03)}{5(0.02)+0.01+5(0.03)}$$

$$\Rightarrow \frac{0.11}{0.11+0.15} \cdot \frac{0.18}{0.13+0.18} \cdot \frac{0.15}{0.11+0.15} \Rightarrow \frac{0.11}{0.26} + \frac{0.18}{0.31} + \frac{0.15}{0.26}$$

$$\Rightarrow (0.4231)(0.5806)(0.5769) \approx 0.142$$

Question 3:

Let $X(t)$ represent the number of cars in a parking lot at time t.

Arrival rate $= \sigma$ per day

Departure rate $= \alpha$ per day

$\{X(t), t \geq 0\}$

state space $S = \{0, 1, 2, 3 \ldots\}$

Index set $t \in [0, \infty)$

a) Infinitesimal Transition Probabilities as $h \to 0$

$P_{ij}(h) = P(X(h+s) = j \mid X(s) = i)$

$\Rightarrow P(X(h) = j \mid X(0) = i)$

$\Rightarrow P(X[T_i] = j;\ T_i \leq h \mid X(0) = i)$

$\Rightarrow P_{ij}(1 - e^{-hv_i})$

$\Rightarrow P_{ij}(hv_i + o(h))$ $\forall$ Small h

$\Rightarrow P_{ij} v_i h + o(h)$

$\Rightarrow q_{ij} h + o(h), j \neq i$

$$P_{ij}(h) = \begin{cases} \sigma h + o(h), & j = i+1 \\ \alpha h i + o(h), & j = i-1 \\ 1 - (\sigma + \alpha i)h + o(h), & j = i \\ o(h), & |i-j| > 1 \end{cases}$$

b) $P_{ij}(t) = P(X(t+s) = j \mid X(s) = i) = P(X(t) = j \mid X(0) = i)$

$$\frac{d}{dt} P_{ij}(t) = \lim_{h \to 0} \frac{P_{ij}(t+h) - P_{ij}(t)}{h}$$

$$\Rightarrow \sum_{k \neq j}^{\infty} q_{kj} P_{ik}(t) - v_j P_{ij}(t)$$

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Question 4:

Let $X(t)$ represent the number of fish remaining at time t.

$\{X(t); t \geq 0\} \longrightarrow$ Pure death CTMC

State space $S = \{0, 1, 2, \ldots N\}$

Index set $t \in [0, \infty)$

death rate $= \alpha$

Birth rate $= \lambda = 0$

(a) $P_{i,i-1} = \dfrac{\alpha i}{\lambda_i + \alpha_i} = \dfrac{i\alpha}{\lambda + i\alpha} = \dfrac{i\alpha}{i\alpha} = 1 \quad \forall\, i = 1, 2, \cdots N$

$P_{i,i+1} = \dfrac{\lambda_i}{\lambda_i + \alpha_i} = \dfrac{\lambda}{\lambda + i\alpha} = \dfrac{0}{i\alpha} = 0$

(b) Let $P_{ij}(t) = P(X(t) = j \mid X(0) = i) \quad \forall\, 0 - \in \{0, 1, 2 \ldots N\}$

$$\frac{d}{dt} P_{ij}(t) = \sum_{k \neq j} q_{kj} P_{ik}(t) - V_j P_{ij}(t)$$

$$\frac{d}{dt} P_{ij}(t) = \alpha(j+1) P_{i,j+1}(t) - \alpha j P_{ij}(t)$$

$$\frac{d}{dt} P_{ij}(t) = \alpha P_{i1}(t)$$

$$P'_{ij}(t) = \alpha P_{i1}(t)$$

The Infinitesimal generator is $\begin{bmatrix} 0 & 0 & 0 \\ \alpha & -\alpha & 0 \\ 0 & 2\alpha & -2\alpha \end{bmatrix}$

(c) $\alpha = 0.003 \quad M = 2$

$S = \{0, 1, 2\} \quad X(0) = 2$

$P(X(t) = k \mid X(0) = n) = ? \quad n, k \in S$

$P(T > t) \quad P_{2,2}(t) = e^{-2\alpha t}$

$P_{2,1}(t) = 2\alpha t e^{-2\alpha t}$

$P_{2,0}(t) = 1 - P_{2,2}(t) - P_{2,1}(t) = 1 - e^{-2\alpha t} - 2\alpha t e^{-2\alpha t}$ {By solving Kolmogorov forward differential Equation

(d) $P(X(5) = 0 \mid X(0) = 2) \Rightarrow P_{2,0}(5) = 1 - e^{-2\alpha 5} - 2\alpha \cdot 5 \cdot e^{-2\alpha \cdot 5}$

Given $\alpha = 0.003$

$2\alpha = 0.006, \quad t = 5$

$e^{-0.03} = 0.9704 \Rightarrow 2\alpha t = 0.03$

$\therefore P_{2,0}(5) = 1 - 0.9704 - (0.03)(0.9704)$

$\Rightarrow 1 - 0.9704 - 0.0291 \Rightarrow 0.0005$

$\boxed{\Rightarrow 0.05\%}$

Question 6!

$\{N_1(t), t \geq 0\}$

$N_1(t)$ represent number of arrivals to the Parking Lot by time t

$N_2(t)$ represent number of departures from the Parking lot by time t

$N_1(0) = 0$

Rate $\lambda = \sigma = 0.05$ car/min

$N_1(t) \sim$ Poisson (λt)

Question 5

a) Let $X(t)$ represent the number of Customers in the system {both waiting & being served} at time t.

$\{X(t); t \geq 0\}$

$S = \{0, 1, 2, \dots\}$

Index Set : $t \in [0, \infty)$

Arrival rate $= \lambda = \frac{1}{4} = 0.25$/min

Service rate $= \mu = 1/5 = 0.2$/min

b) $P_{i,i+1} = \frac{\lambda_i}{\lambda_i + \mu_i} = \frac{\lambda}{\lambda + \mu_i}$ Let $\lambda + \mu_i = v_i$

$\approx \frac{\lambda}{v_i}$

$P_{i,i-1} = \frac{\mu_i}{\lambda_i + \mu_i} = \frac{\mu_i}{v_i}$ $P_{ii} = 0$, $P_{ij} = 0$

c) $P_{23} \cdot P_{34} \cdot P_{43} \cdot P_{32} = \frac{\lambda}{\lambda + 2\mu} \cdot \frac{\lambda}{\lambda + 3\mu} \cdot \frac{\mu_4}{\lambda + \mu_4} \cdot \frac{\mu_3}{\lambda + \mu_3}$

$\Rightarrow \frac{0.25}{0.65} \cdot \frac{0.25}{0.85} \cdot \frac{0.8}{1.05} \cdot \frac{0.6}{0.85}$

$\Rightarrow 0.3846 \cdot 0.2941 \cdot 0.7619 \cdot 0.7059$

$\Rightarrow 0.0604$

d) Infinitesimal Transition Probabilities (as $h \to 0$)

$\Rightarrow P(X(t+h) = i+1 \mid X(t) = i) = \lambda h + o(h) = 0.25h + o(h)$

$\Rightarrow P(X(t+h) = i-1 \mid X(t) = i) = \mu_i h + o(h) = \min(i, 5) \cdot 0.2h + o(h)$

$\Rightarrow P(X(t+h) = i \mid X(t) = i) = 1 - (\lambda + \mu_i)h + o(h)$

$$P_{ij} = \begin{cases} 0.25h + o(h), & j = i+1 \\ \min(i,5) \cdot 0.2h + o(h), & j = i-1 \\ 1 - (\lambda + \mu_i)h + o(h) \\ o(h), & |i-j| > 1 \end{cases}$$

e) Without solving! Kolmogorov differential Equations!

Let $P_{ij}(t) = P(X(t) = j \mid X(0) = i)$

$P'_{ij}(t) = \sum_{k \neq j} P_{ik}(t) \cdot q_{kj} - P_{ij}(t) \cdot v_j$

Where: q_{kj} is the transition rate from state $k \to j$

$v_j = \sum_{i \neq j} q_{ji}$

$P'_{ij}(t) = P_{i,j-1}(t) \cdot 0.25 + P_{i,j+1}(t) \cdot 1.0 - P_{ij}(t) \cdot 1.25$

Question 6:

$\{N_1(t), t \geq 0\}$

$N_1(t)$ represent number of arrivals to the parking lot by time t

$N_2(t)$ represent number of departures from the parking lot by time t

$N_1(0) = 0$

Rate $\lambda = \alpha = 0.05$ car/min

$N_1(t) \sim$ Poisson (λt)

$N_1(t+s) - N_1(s) \sim$ Poisson (λt)

$\{N_2(t), t \geq 0\}$

$N_2(0) = 0$

Rate $\mu = \alpha = 0.07$ car/min

$N_2(t) \sim$ Poisson (μt)

$N_2(t+s) - N(s) \sim$ Poisson (μt)

ⓐ The interarrival time in a poisson process are exponentially distributed.

Arrival are exponential with rate λ, the sum of 5 independent exponential (λ)

$$\text{Gamma}\left(K=5, \theta = \frac{1}{\lambda}\right)$$

$$E[T_5] = \frac{5}{\lambda} = \frac{5}{0.05} = 100 \text{days}$$

ⓑ $P(T_5 < 5 \mid T_2 > 3) = ?$

Let $T_n \sim$ Gamma(n, λ)

$$P(T_5 < 5 \mid T_2 > 3) = \frac{P(T_2 > 3, T_5 < 5)}{P(T_2 > 3)}$$

Using R to evaluate

t2_min <- min_to_days(3)

t5_max <- min_to_days(5)

$$\Rightarrow \frac{P(T_2 > 3, T_5 < 5)}{P(T_2 > 3)} = \frac{\text{Pgamma}(t5\_max, \text{shape}=5, \text{rate}=\text{delta}) - \text{Pgamma}(t2\_min, \text{shape}=5, \text{rate}=\text{delta})}{1 - \text{pgamma}(t2\_min, \text{shape}=2, \text{rate}=\text{delta})}$$

$\Rightarrow 1.211952e^{-21}$

ⓒ $P(2 \leq T_5 \leq 5)$?

$T_5^{\text{leave}} \sim$ Gamma$(5, \alpha)$

Convert α to per minute! $\alpha = \frac{0.07}{1440} \simeq 4.86 \times 10^{-5}$

$P(2 < T_5 < 5) = F_{T_5}(5) - F_{T_5}(2)$ Using R $F_{T_5}(t) =$ Pgamma $(t, 5, 4.86 \times 10^{-5})$

$\Rightarrow$ Pgamma$\left(5, \frac{1}{0.07}\right)$ − Pgamma$\left(\alpha = \overset{\text{to_days}}{5}, \beta = \frac{1}{0.07}\right)$

$\Rightarrow 6.995003e^{-21}$

D) $P(2 < T_2^{leave} < 3 \mid T_5^{leave} < 6) = ?$

$$P(2 < T_2 < 3 \mid T_5 < 6) = \frac{P(2 < T_2 < 3,\ T_5 < 6)}{P(T_5 < 6)}$$

$T_2 \sim Gamma(2, \alpha)$

$T_5 \sim Gamma(5, \alpha)$

Using R:

```
t2a <- min_to_days(2)
t2b <- min_to_days(3)
t5max <- min_to_days(6)
```

$$P(2<T_2<3 \mid T_5<6) = \frac{(\text{Pgamma}(t2b, shape=2, rate=alpha) - \text{Pgamma}(t2a, shape=2, rate=alpha)) * \text{Pgamma}(t5max, shape=3, rate=alpha)}{\text{Pgamma}(t5max, shape=5, rate=alpha)}$$

$P(2<T_2<3 \mid T_5<6) = 1.388752$

E) what is the standard deviation of the waiting time until the 5^{th} car arrives?

$T_5^{arrive} \sim Gamma(5, \lambda)$

$$SD(T_n) = \frac{\sqrt{n}}{\lambda} \Rightarrow \frac{\sqrt{5}}{0.05} = \frac{2.236}{0.05} = 44.72 \text{ days}$$

F) Sample mean $(\bar{T}) = \frac{1}{6}\sum_{i=1}^{6} T_i = E[\bar{T}] = \frac{1}{\mu} = \frac{1}{0.07} = 14.29$ days

$$Var[\bar{T}] = \frac{\frac{1}{\mu^2}}{6} = \frac{1}{6\mu^2} = \frac{1}{6(0.07)^2} = 34 \text{ days}$$

Expected Sample Var $= E[S^2] = \frac{1}{\mu^2} = \frac{1}{\mu^2} = 204$ days

I) $T_1, \ldots, T_{30} \sim Exp(\alpha)$

$E[\bar{T}_{30}] = \frac{1}{\alpha} = \frac{1}{0.07} = 14.29$ days

Variance of Sample mean : $V[\bar{T}_{30}] = \frac{1}{30\alpha^2} = \frac{1}{30 * 0.07^2} = \frac{1}{0.147} = 6.80$

SD of Sample mean $\approx \sqrt{6.80} = 2.61$ days.